RADHIKA GUPTA

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EDUCATION

Ajay Kumar Garg Engineering College — **Ghaziabad, Uttar Pradesh**
B.Tech - Computer Science — *CGPA: 8.5/10 | 2024 - 2028*

Heritage Academy — **Ghaziabad, Uttar Pradesh**
Senior Secondary (12th), CBSE — *Percentage: 94% | 2024*

Heritage Academy — **Ghaziabad, Uttar Pradesh**
Secondary (10th), CBSE — *Percentage: 95.4% | 2022*

TECHNICAL SKILLS

Programming Languages: C++, JavaScript, Python
Frontend Technologies: ReactJs, HTML5, CSS3, Tailwind CSS
Backend Technologies: NodeJs, ExpressJs, Socket.io, REST APIs, Mongoose
Databases: MongoDB, SQL, Firebase
Core CS Fundamentals: Data Structures and Algorithms, Operating System, Database Management System, Computer Networks, Object Oriented Programming
Tools and Others: Git, GitHub, Postman, Swagger, Vercel, Render, Cloudinary

PROJECTS

Wave — GitHub | Live Demo

React, Node.js, Express.js, MongoDB, Socket.io, JWT, ZegoCloud, Google OAuth, Tailwind CSS

- Engineered a full-stack real-time messaging platform supporting one-on-one and group conversations, implementing live typing indicators, read receipts, online presence tracking, and unread message synchronization through a persistent Socket.io connection layered on a modular Express REST API.
- Architected a per-user socket room delivery system to guarantee reliable real-time message, read-receipt, and presence updates regardless of which conversation a recipient had active, resolving a class of missed-delivery bugs inherent to naive conversation-only room broadcasting.
- Integrated ZegoCloud's Call Kit and ZIM signaling SDK to deliver in-app voice and video calling with live call invitations, and designed a call-outcome logging pipeline that persists missed, declined, and completed calls as first-class chat messages with duration tracking.
- Built group chat functionality with role-based admin permissions, member management, and leave/removal workflows, alongside message-level features including emoji reactions, in-place editing, soft deletion, and multi-format (image/video/audio/document) attachment handling via Cloudinary.

GeoTime — GitHub | Live Demo

React, Vite, Tailwind CSS, Mapbox GL JS, Node.js, Express.js, MongoDB, Sentinel Hub, NASA GIBS, NASA FIRMS, Gemini API, jsPDF

- Developed a full-stack geospatial intelligence platform for exploring historical satellite imagery and monitoring environmental changes, including vegetation health, deforestation, urban expansion, and fire activity.
- Built an interactive Mapbox GL JS interface with date-based imagery selection and swipe comparison, enabling users to examine satellite observations across different time periods.
- Integrated NDVI, NDWI, and NDBI spectral indices with NASA GIBS, Copernicus Sentinel, and NASA FIRMS data to support vegetation, water-related, built-up-area, and fire-hotspot analysis.
- Implemented GeoMate, a Gemini-powered geospatial assistant, and automated research report generation with jsPDF-powered PDF and CSV exports to help users interpret and document findings.

DoodleDoc — GitHub | Live Demo

React, Konva, Zustand, Tailwind CSS

- Engineered a canvas-based whiteboard application from scratch, implementing freehand drawing, shape rendering, text editing, object transformations, and image export using React-Konva and custom event-driven architecture.
- Designed and developed a snapshot-based Undo/Redo system with persistent room-specific storage, enabling reliable state recovery and seamless restoration of user work across sessions.
- Built an interactive editing framework supporting object selection, drag-and-drop, resizing, formatting controls, mobile touch interactions, and centralized state management through Zustand, delivering a responsive cross-device user experience.

ACHIEVEMENTS

- Solved **450+** DSA problems on GeeksforGeeks.
- Solved **300+** DSA problems on Leetcode.

CODING PROFILES

LeetCode | GeeksforGeeks | GitHub